

Assignment: Smart Data Quality Validation on Databricks

Corporate Training Assignment
Data Quality · Anomaly Detection · AI-Assisted Rule Generation · Quality Scoring
Suggested Duration: 60–90 Minutes | Difficulty: Intermediate

Objective

You are working as a Data Engineer for an e-commerce company receiving customer and transaction data from multiple sources. Build a Smart Data Quality Validation Framework using Databricks, SQL/PySpark, and optionally Databricks AI functions.

Dataset

Column	Description
customer_id	Unique customer identifier
customer_name	Customer name
email	Customer email address
country	Two-letter country code
signup_date	Customer registration date
transaction_id	Unique transaction identifier
transaction_amount	Transaction amount
transaction_date	Date of transaction

Introduce bad-quality records: missing customer IDs, duplicate transaction IDs, invalid emails/countries, negative or extreme transaction amounts, stale dates, and NULL customer names.

Task 1 — Data Profiling

Calculate total records, NULL counts, distinct customers, duplicate transaction IDs, and minimum, maximum, and average transaction amounts. Create a profiling summary.

Task 2 — Traditional Data Quality Rules

Implement at least five deterministic rules: customer_id NOT NULL; transaction_id UNIQUE; valid email pattern; transaction_amount > 0; and approved country codes (IN, US, UK, AE, SG). Add pass/fail validation columns and create VALID_RECORDS and INVALID_RECORDS outputs.

Task 3 — Anomaly Detection

Use IQR or MAD to detect anomalies in transaction_amount. Report the anomaly count and highest anomalous amount. Explain whether every statistical anomaly necessarily represents bad data.

Task 4 — AI-Generated Rules (Optional Advanced)

Use a Databricks Foundation Model or AI function to analyze schema and sample records and propose structured JSON validation rules. Review each rule as APPROVE, MODIFY, or REJECT. Explain why human review is required.

Task 5 — Data Quality Score

Calculate scores for Completeness, Validity, Uniqueness, Consistency, Accuracy, and Timeliness. Produce an overall 0–100 score. Bonus: weight Critical rules as 3 and Warning rules as 1.

Final Deliverables

- Databricks Notebook with SQL/PySpark code.
- Data Profiling Report.
- Valid and invalid record results.
- Anomaly Detection Results.
- Dimension-level and Overall Data Quality Score.

6. AI-generated rules, if attempted.
7. A 5–10 line conclusion.

Evaluation Criteria

Data Profiling: 15% · Validation Rules: 25% · Anomaly Detection: 20% · AI Rule Generation: 15% · Quality Scoring: 20% · Conclusion: 5%

Key Principle

AI proposes; deterministic code disposes. Use AI to accelerate rule creation and anomaly explanation while keeping production validation deterministic, testable, auditable, and reproducible.